

SnehoAyu

স্নেহ আয়ু

Preterm Infant Care Companion — mHealth PWA

PRODUCT REQUIREMENTS DOCUMENT

Version 2.0 | June 2025 | Final Draft

Prepared for: Purnima Chakraborty

Confidential — For Client, Developer & Partner Review

Study: "A Study to Evaluate the Effectiveness of a Mobile-Based M-Health Intervention (SnehoAyu App) on Clinical Outcome of Preterm Infants, Knowledge on Newborn Care, Mental Well-Being, and Self-Efficacy among Mothers at Selected Hospitals of West Bengal"

Document Control

Field	Detail
Product Name	SnehoAyu (স্নেহ আয়ু)
Document Type	Product Requirements Document (PRD)
Version	2.0 — Complete Edition
Date	June 2025
Researcher	Purnima Chakrabortty
Study Type	Randomised Controlled Trial (RCT)
Platform	Progressive Web App (PWA) — Android-first
Languages	Bengali (primary), Hindi, English
Study Sites	Bankura MCH, Burdwan MCH, Purba Bardhaman (Pilot: Deban Mahato MCH, Purulia)
Sample Size	272 mothers (136 study group + 136 control group)
Study Duration	6 months post-discharge per participant
Document Status	Final Draft — Approved for Development

1. Executive Summary

SnehoAyu is both a research instrument and a care companion. It is a Progressive Web App (PWA) built to support mothers of NICU-discharged preterm infants — and simultaneously to generate measurable, peer-reviewable evidence that such digital support improves infant outcomes, maternal knowledge, mental well-being, and self-efficacy.

India contributes over 20% of global preterm births. In West Bengal alone, the preterm birth prevalence is 16% — nearly 1 in 6 deliveries. Once a preterm infant is discharged from the NICU, families receive verbal guidance and a printed sheet. There is no structured system for daily monitoring, reminder support, danger-sign guidance, or emotional care at home. The result: high readmission rates, growth faltering, missed vaccinations, and maternal anxiety.

SnehoAyu closes this gap with a single digital companion. It is distributed through hospitals at discharge, free to parents, and requires only an Android smartphone. It works offline. It speaks Bengali.

This document is the complete specification for what must be built — covering all screens, features, data collection instruments, research workflows, technical constraints, and the phased delivery plan.

2. The Research Study

Understanding the research context is essential for every developer who works on this app. SnehoAyu is not a standalone commercial product — it is the intervention arm of a formal Randomised Controlled Trial (RCT). The app must therefore serve two roles simultaneously: a daily care tool for mothers, and a data collection platform for the researcher.

2.1 Study Design

Parameter	Detail
Design	True Experimental — Randomised Pretest–Posttest Control Group Design
Approach	Quantitative
Groups	Study Group (app) + Control Group (standard care only)
Randomisation	Stratified by birth weight: <1500g, 1500–2500g, >2500g. Lottery method.
Blinding	Data analysis blinded (done by statistician)
Control Group Rule	Control group data collection completed BEFORE study group begins
Intervention Period	6 months post-NICU discharge per participant
Follow-up Points	Baseline (pre-discharge), 1 month, 3 months, 6 months post-discharge

2.2 Study Sites

Role	Hospital
Primary Study Sites	Bankura Medical College & Hospital (Bankura)
	Burdwan Medical College & Hospital (Burdwan)
	Purba Bardhaman (NICU site)
Pilot Site	Deban Mahato Medical College & Hospital (Purulia)

2.3 Measurement Schedule

Data Collection Point	When	What is Collected
Baseline (O1)	Before NICU discharge (app installed same day)	Demographic data, infant profile, baseline growth, baseline knowledge (MCQ), baseline WHO-5, baseline PSOC
Post-test 1 (O2)	1 month after discharge	Growth, development (TDSC), immunization, breastfeeding, knowledge, WHO-5, PSOC
Post-test 2 (O3)	3 months after discharge	Same as O2
Post-test 3 (O4)	6 months after discharge	Same as O2 + readmission record + full outcome summary

2.4 Inclusion and Exclusion Criteria

Inclusion

- Mothers of preterm infants (born before 37 completed weeks)
- Willing to participate and available during study period
- Can read and write Bengali
- Owns or has access to an Android smartphone
- Infant admitted to NICU for at least 14 days and stable at discharge
- Infant on at least katori feeding at time of discharge
- Primigravida mother; Normal Delivery

Exclusion

- Infants with major congenital anomalies or terminal illness
- Mothers who are or have a close family member who is a health professional
- Mothers with any acute illness at time of enrolment

3. Users & Roles

3.1 Role Overview

Role	Who	App Access
Mother / Caregiver	Primary participant. Mothers of NICU-discharged preterm infants.	Full parent-facing app — all care, tracking, learning, and support features
Researcher / Nurse	Purnima Chakrabortty and designated data collectors at each hospital.	Admin panel — enrol participants, view submitted data, manage app content, trigger follow-up reminders
Hospital Staff	NICU nurses who install and demonstrate the app at discharge.	Onboarding flow only — link mother to study group, generate participant code

3.2 Primary User Persona — The Mother

Attribute	Detail
Name (representative)	Priya Devi
Age	22–32 years, first-time mother
Location	Semi-urban or rural West Bengal
Language	Bengali — primary. Limited English/Hindi.
Device	Mid-range Android smartphone. 4G. Limited storage.
Digital literacy	Uses WhatsApp and YouTube. Not comfortable with complex apps.
Situation	Baby born 30–34 weeks. Spent 3 weeks in NICU. Discharged home.
Biggest fear	"My baby will fall sick again and I won't know what to do."
Primary need	Simple daily guidance she can follow and refer back to, in Bengali.

4. What We Are Building

SnehoAyu is a single Progressive Web App with two distinct interfaces: (1) a Mother App — the daily care companion parents use at home, and (2) a Researcher Panel — the backend interface Purnima Chakraborty uses to manage the study, enrol participants, and access collected data. Both must be built in Phase 1.

4.1 Platform

- Progressive Web App (PWA) — installable from the browser, no Play Store required
- Android Chrome primary target; must also work on Samsung Internet
- Offline-first — core features must work without internet (checklist, danger signs, saved content)
- Service Worker for caching and background sync when connection restored
- Responsive design — mobile-only layout (360px–480px viewport target)

4.2 Languages

- Bengali (বাংলা) — primary language, mandatory for all UI strings and content
- Hindi — secondary, for broader reach post-study
- English — tertiary, for researcher panel and technical use
- Language preference saved per user account
- Default language: Bengali at first launch

4.3 In Scope — Phase 1 (This Document)

- Complete parent-facing PWA with all care and tracking features
- All 5 research measurement tools embedded as in-app forms
- Researcher / admin panel for study management and data access
- Daily SMS text messages and weekly audio message delivery
- Telehealth video call capability (researcher to mother)
- Real authentication with OTP
- Full Bengali UI
- Offline mode

4.4 Out of Scope

- Native Android or iOS app (PWA is the target)
- Integration with government HMIS / immunization databases
- Multi-baby support (twins, siblings — not required for this study)
- Payment or subscription billing
- Clinical decision support or prescription tools
- Physical device integration (smart scales, oximeters — data entered manually)

5. Mother App — Feature Specifications

This section describes every screen and feature the mother sees. For each feature: what it does, why it matters for the research, what exists in the current codebase, and what still needs to be built.

5.1 Onboarding & Authentication

UI: Partially Complete | Backend: Not Started | Priority: P0 — Blocker

What It Does

Allows a mother to create her account and link to the study at the time of NICU discharge. The nurse installs the app and creates the account with the mother before she leaves the hospital.

Onboarding Flow (Step by Step)

1. Nurse opens SnehoAyu PWA on mother's phone and taps 'Enrol New Participant'
2. Nurse enters the hospital code (pre-assigned to each study site)
3. Nurse enters mother's phone number — OTP sent via SMS
4. Mother verifies OTP and sets a simple 4-digit PIN (no password typing)
5. Mother's profile form: name, age, education, religion, income class, residence, family type
6. Infant profile form: name, birth date, gestational age, birth weight, sex, NICU duration, KMC, initial feeding method
7. Mother assigned a participant code (anonymised) — linked to study group or control group
8. App language set to Bengali by default. Option to change.
9. 2-hour training session conducted by nurse over 3 days before discharge
10. Within 7 days of discharge: phone follow-up call by researcher to confirm app is being used

Login / Re-entry

- Returning login: phone number + 4-digit PIN
- PIN reset: OTP to registered phone number
- Session stays active for 30 days — mother does not have to log in daily

What Needs to Be Built

- OTP-based phone authentication (SMS gateway — suggest MSG91 or Twilio India)
- 4-digit PIN setup and login
- Mother profile form (all demographic fields from Tool I of the proposal)
- Infant profile form (all infant fields from Tool I)
- Hospital code linking — connects mother to study site and researcher
- Participant code assignment (study vs control group randomisation handled by researcher panel)
- Session persistence (30-day active session)

5.2 Home Dashboard

UI: Complete | Backend: Not Started | Priority: P0

What It Does

The first screen after login. A single glance at today's care status, what tasks are done, what needs attention, and what reminder is upcoming. Designed to be understood in under 5 seconds by a sleep-deprived mother.

Components

- Baby status card — baby name, current age in days/weeks, adjusted age, weight from last reading
- Today's care ring — circular progress showing % of daily checklist completed
- Quick action buttons — Log Feeding, Log Temperature, Log KMC (one-tap data entry)
- Feeding summary — feeds completed today vs target
- Health stats strip — last recorded temperature, weight, SpO2
- Next vaccination reminder card — vaccine name, due date, days remaining
- Daily message card — shows today's SMS-format care tip (displayed in-app too)

What Needs to Be Built

- All home cards wired to live backend data
- Quick action buttons that write readings to backend
- Dynamic vaccination countdown from immunization module

5.3 Daily Care Checklist

UI: Complete | Backend: Not Started | Priority: P0

What It Does

A structured daily checklist of care tasks. The mother ticks each item as done. The checklist resets every day. Completion data feeds into the research outcomes and the home dashboard progress ring.

Default Daily Tasks

- Breastfeeding — number of feeds and volume (8–12 times per 24 hours)
- Kangaroo Mother Care (KMC) — minimum 1–2 hours logged
- Temperature check — morning and evening (normal 36.5–37.5°C)
- Weight check — on applicable days (weekly or as advised)
- Skin and cord care
- Medication — if prescribed (customisable field)
- Danger signs self-check — prompt to review once per day

What Needs to Be Built

- Backend persistence — one checklist record per day per participant
- Daily reset at midnight IST via server-side cron
- 7-day and 30-day history view — mother can see past days
- Completion rate calculation exposed to researcher panel

5.4 Growth Tracking

UI: Partially Complete (no chart) | Backend: Not Started | Priority: P1

What It Does

Records and displays the baby's weight, length, and head circumference over time. Compared against WHO growth chart standards adjusted for the baby's corrected (gestational) age. This is a core research outcome — growth data is collected at baseline, 1 month, 3 months, and 6 months.

Components

- Current measurements card — weight (g), length (cm), head circumference (cm)
- WHO z-score display — weight-for-age, length-for-age, HC-for-age
- Growth chart — line chart plotting all readings against WHO percentile bands (3rd, 15th, 50th, 85th, 97th)
- Readings history table — all past entries with date and who entered it
- Manual entry form — mother or nurse enters new reading
- Alert flag — if weight drops or z-score falls below -2 SD (flagged for researcher)

Research Note

Growth data must use CORRECTED age (chronological age minus weeks of prematurity) for WHO chart comparison. This calculation must be automatic in the backend.

What Needs to Be Built

- Backend API to store and retrieve growth readings with timestamps
- WHO growth chart data (boys and girls, weight/length/HC, 0–6 months corrected age)
- Z-score calculation engine
- Chart rendering component (suggest Recharts or Chart.js)
- Manual entry form with validation
- Alert system for z-score < -2 SD

5.5 Developmental Milestone Tracker (TDSC)

UI: Not Built | Backend: Not Started | Priority: P1

What It Does

Tracks developmental milestones using the Trivandrum Developmental Screening Chart (TDSC). This is a pass/fail screening tool used at 1, 3, and 6 month follow-ups. Failure on even one item to the left of the scale indicates suspected developmental delay.

TDSC Items to Be Implemented (18 items, 0–25 months)

Item	Developmental Task	Lower Limit	Upper Limit
1	Social smile	1 day	2 months
2	Eyes follow pen/pencil	1m 3d	3m
3	Holds head steady	1m 3d	3m 24d
4	Rolls from back to stomach	2m 21d	4m 24d
5	Turns head to sound of bell	3m	5m 24d
6	Transfer objects hand to hand	4m 3d	7m
7	Raises self to sitting	5m 24d	11m
8	Standing up by furniture	6m 9d	11m
9	Fine prehension (pellet)	6m 24d	11m
10	Pat-a-cake	6m 24d	12m 21d
11	Walks with help	7m 24d	13m
12	Throws ball	9m 15d	16m 24d
13	Walks alone	9m 27d	17m 12d
14	Says two words	11m 6d	19m
15	Walk backwards	11m 6d	19m 15d
16	Walk upstairs with help	12m 6d	24m 15d
17	Points to 3 parts of doll	15m 9d	24m 15d
18	Remove garments	21m	25m

Screening Logic

- App calculates corrected age and shows only the items relevant to that age range
- Each item: Pass ✓ / Fail ✗
- If any item fails that should have been passed by corrected age → flag as 'Suspected Delay'
- Result saved per assessment date and exposed in researcher panel

What Needs to Be Built

- TDSC screen with item-by-item pass/fail interface

- Corrected age engine to filter relevant items
- Delay detection logic
- Assessment history view

5.6 Immunization Schedule & Tracker

UI: Complete | Backend: Not Started | Priority: P1

What It Does

Displays the baby's complete vaccination schedule generated from birth date. Preterm infants follow the vaccination schedule based on chronological (birth) age, not corrected age. Tracks which vaccines are done and sends reminders for upcoming ones.

Components

- Full vaccine schedule — all vaccines from birth to 6 months with due dates
- Completed vaccines — marked with date given, batch number (optional), given by
- Pending vaccines — sorted by due date with days remaining
- Progress bar — percentage of 6-month schedule completed
- Reminder system — in-app alert 7 days and 1 day before due date
- Vaccine detail card — what the vaccine protects against, common side effects
- Immunization status for researcher — up to date / delayed / not started

What Needs to Be Built

- Schedule generation engine from birth date (IAP 2023 schedule)
- Mark-as-done form with date and optional batch number
- Reminder notification via PWA push notification
- Status export for researcher panel

5.7 Breastfeeding Log & Assessment

UI: Not Built | Backend: Not Started | Priority: P1

What It Does

Logs daily breastfeeding data and assesses feeding patterns against the structured interview schedule from the research tool (Tool II, Section B4). This is a core research outcome measured at each follow-up.

Daily Log

- Feed type: Exclusive breastfeeding / Predominantly breastfeeding / Mixed / Formula only
- Number of feeds in 24 hours
- Duration of a typical feed session
- Night feeds count
- Feeding cues practiced (yes/always / sometimes / no — fixed schedule)

Feeding Assessment Form (Research Tool — at 1, 3, 6 months)

At each research follow-up, the full 8-item breastfeeding interview schedule is presented in-app:

- Question 1: Currently breastfeeding? (Exclusive / Predominantly / Mixed / Not breastfeeding)
- Question 2: Reason if not exclusive
- Question 3: Frequency per 24 hours
- Question 4: Duration of sessions
- Question 5: Night feeding frequency
- Question 6: Feeding on cues
- Question 7: Feeding problems (multi-select)
- Question 8: Expressed breast milk used?
- Question 9: Alternative feeding methods used?

Scoring: 24–28 Excellent | 18–23 Good | 12–17 Fair | ≤11 Poor

What Needs to Be Built

- Daily breastfeeding quick log (from Home or Checklist)
- Full assessment form (locked to follow-up dates)
- Automated scoring and grading
- History and trend display

5.8 Danger Signs Guide

UI: Complete | Backend: Not Started | Priority: P1

What It Does

A quick-reference guide for the mother to identify warning signs that need immediate medical attention. Expandable cards with symptoms, what to observe, and what to do. One tap calls the hospital helpline.

Danger Signs Covered

- Breathing difficulty or fast breathing (>60 breaths/min)
- Blue lips or fingertips (cyanosis)
- Temperature below 36.5°C or above 38°C
- Refusing to feed or unable to feed
- Lethargic, floppy, or unresponsive
- Convulsions or seizures
- Jaundice spreading below the navel
- Sunken fontanelle
- Significant weight loss or no weight gain
- Redness or discharge from umbilical cord stump

What Needs to Be Built

- Emergency call button wired to hospital helpline (configured per hospital in admin panel)
- Bengali translation of all danger sign descriptions
- Deep-link from Learning Hub

5.9 Knowledge Assessment — 15-Question MCQ (Research Tool III)

UI: Not Built | Backend: Not Started | Priority: P0 — Research Instrument

What It Does

The 15-question multiple-choice knowledge test from Tool III of the research proposal. Administered at baseline and all three follow-up points. This is a PRIMARY research outcome.

This is not a quiz for fun — it is a validated research instrument. The questions and answer options must be implemented EXACTLY as specified in the proposal. No rewording.

The 15 Questions

Q	Topic	Correct Answer
1	How to know baby gets enough milk	Baby pees >6 times/day
2	How to keep preterm baby warm	Kangaroo Care (skin-to-skin)
3	Baby suddenly refuses to feed	Could be an emergency
4	Duration of exclusive breastfeeding	6 months
5	Which is NOT a danger sign	Baby pees 8–10 times/day
6	When to start breastfeeding after birth	Within 1 hour
7	Definition of preterm birth	Born before 8 months (37 weeks)
8	Who advises vaccination schedule	Doctor or health worker
9	Umbilical cord care	Keep clean and dry
10	Daily KMC duration	1–2 hours minimum
11	Keeping baby warm at home	Skin-to-skin contact
12	Major risk for preterm babies	Hypothermia (low body temp)
13	Why preterm babies stay longer	Not learned to feed/breathe
14	Baby vomits often — what to do	Consult a doctor
15	Baby feels very hot — what to do	Consult a doctor

Scoring

- 1 mark per correct answer. Total: 15 marks.
- Poor Knowledge: 0–5 (0–33%)
- Moderate Knowledge: 6–10 (34–66%)

- Good Knowledge: 11–15 (73–100%)

Administration Rules

- Presented as interview-style form (nurse asks, mother answers — or mother self-completes if literate)
- Administered at: Baseline, 1 month, 3 months, 6 months
- All 4 scores stored per participant and exported in researcher panel
- Form locked after submission at each time point — cannot be re-taken

5.10 Mental Well-Being Assessment — WHO-5 Index (Research Tool IV)

UI: Not Built | Backend: Not Started | Priority: P0 — Research Instrument

What It Does

The validated WHO-5 Well-Being Index. Measures the mother's mental well-being over the past two weeks. This is a PRIMARY research outcome administered at all four time points.

The 5 Questions

#	Statement	Scale
1	I have felt cheerful and in good spirits	0 (At no time) → 5 (All of the time)
2	I have felt calm and relaxed	0 → 5
3	I have felt active and vigorous	0 → 5
4	I woke up feeling fresh and rested	0 → 5
5	My daily life has been filled with interesting things	0 → 5

Scoring

- Raw score: sum of all 5 answers (range 0–25)
- Percentage score: raw score × 4 (range 0–100)
- Score < 50% (raw < 13): Poor mental well-being — flag for researcher attention
- Score ≥ 50%: Good mental well-being

Implementation Rules

- 6-point radio scale per question with Bengali labels
- Score auto-calculated and displayed to mother with simple interpretation
- Administered at: Baseline, 1 month, 3 months, 6 months
- Full score history stored and exported to researcher panel
- If score < 50%: gentle in-app message encouraging her to contact the researcher

5.11 Self-Efficacy Assessment — PSOC Scale (Research Tool V)

UI: Not Built | Backend: Not Started | Priority: P0 — Research Instrument

What It Does

The Parenting Sense of Competence Scale (PSOC), developed by Gibaud-Wallston & Wandersman (1978). Measures the mother's self-efficacy and satisfaction in caring for her preterm infant. PRIMARY research outcome, administered at all four time points.

The 17 Items

Item	Statement (Summary)	Scoring
1	Problems of child care are easy once you understand your child	Direct
2	Frustrated now while child is at this age	Reverse
3	Go to bed same as wake up — not accomplished much	Reverse
4	Feel like I'm being manipulated when I should be in control	Reverse
5	My mother was better prepared than I am	Reverse
6	I would be a fine model for a new mother	Direct
7	Being a parent is manageable, problems easily solved	Direct
8	Difficult not knowing if you're doing a good or bad job	Reverse
9	Sometimes I feel I'm not getting anything done	Reverse
10	I meet my personal expectations for caring for my child	Direct
11	If anyone can find the answer for my child, I am the one	Direct
12	My talents and interests are in other areas, not parenting	Reverse
13	I feel thoroughly familiar with this role	Direct
14	If parenting were more interesting, I'd be more motivated	Reverse
15	I have all the skills necessary to be a good mother	Direct
16	Being a parent makes me tense and anxious	Reverse
17	Being a good mother is a reward in itself	Direct

Scoring

- 6-point Likert scale: 1 = Strongly Disagree → 6 = Strongly Agree
- Reverse-scored items: 2, 3, 4, 5, 8, 9, 12, 14, 16 (score becomes: 6→1, 5→2, 4→3, 3→4, 2→5, 1→6)
- Two subscales: Efficacy (items 1,6,7,10,11,13,15,17) | Satisfaction (items 2,3,4,5,8,9,12,14,16)
- Grading: High $\geq$ Median + 1SD | Average = Median $\pm$ 1SD | Low $\leq$ Median – 1SD

Implementation Rules

- Presented as 17-item 6-point scale in Bengali
- Reverse scoring handled automatically in backend — mother selects naturally
- Both subscale scores and total score stored
- Administered at: Baseline, 1 month, 3 months, 6 months
- Full score history stored and exported

5.12 Learning Hub

UI: Partially Complete | Backend: Not Started | Priority: P1

What It Does

A library of care guidance content for mothers. Short articles and videos on feeding, KMC, growth, danger signs, and emotional support. Available in Bengali (primary), Hindi, and English.

Content Categories

- Feeding & Breastfeeding — latching, expressed milk, feeding cues, when to worry
- Kangaroo Mother Care (KMC) — how to, duration, benefits
- Growth & Development — what to expect, how to measure, adjusted age
- Danger Signs — what to watch for and when to call (links to Danger Signs screen)
- Emotional Support — coping with anxiety, bonding, asking for help
- Immunization — what each vaccine is for, schedule, side effects

Content Requirements

- All content medically reviewed and approved by Purnima Chakraborty before launch
- Maximum 3 minutes to read or watch per card
- Available in Bengali — all text and audio narration
- Culturally appropriate imagery
- Weekly spotlight card — rotates to highlight a timely topic

What Needs to Be Built

- Content database / CMS (even a simple JSON-based CMS is acceptable for Phase 1)
- Language toggle — Bengali / Hindi / English per user preference
- Content detail view when card is tapped
- Read/viewed tracking per participant (research engagement metric)

5.13 Daily SMS & Weekly Audio Messages

UI: Not Built | Backend: Not Started | Priority: P1 — Mandated by Study

This feature is explicitly required by the research proposal: 'Send a text message daily and an audio message weekly.' It is not optional.

Daily SMS / Text Message

- One care tip or reminder sent to mother's phone every day via SMS
- Message content: rotation of care tips, check reminders, motivational messages
- All messages in Bengali
- Also displayed as a card on the Home Dashboard (in-app version)
- Message log — mother can scroll back to see past messages in-app
- SMS gateway: MSG91 or similar India-based provider

Weekly Audio Message

- One audio message sent weekly — approximately 2–3 minutes
- Content: weekly care topic, recorded by Purnima Chakraborty or a trusted nurse voice
- Delivered as an in-app notification with audio player
- Stored in-app so playable offline
- Topics follow the study timeline (Week 1: KMC basics, Week 2: Feeding cues, etc.)

What Needs to Be Built

- SMS gateway integration for daily text messages
- Message content library (minimum 180 unique daily messages for 6-month period)
- Weekly audio message file hosting and delivery
- In-app notification for weekly audio
- Message history screen in app

5.14 Telehealth — Video Call with Researcher

UI: Not Built | Backend: Not Started | Priority: P1 — Mandated by Study

Explicitly required by the proposal: 'Weekly video call to each participant if needed' and 'Video calls with parents if needed — provide ongoing guidance.'

What It Does

Allows Purnima Chakraborty (or a designated researcher) to initiate a video call with a study participant directly from the researcher panel. The mother receives a push notification and taps to join.

Functionality

- Researcher initiates call from admin panel — selects participant, taps 'Video Call'
- Mother receives PWA push notification: 'Purnima Chakraborty is calling. Tap to join.'
- Mother taps notification — opens in-app video call screen

- Call ends when either party disconnects
- Call log stored: date, time, duration, participant code

Technical Implementation Options

- Option A: Daily.co or Whereby — embeddable video SDK, free tier available
- Option B: Jitsi Meet — open source, self-hostable, no per-minute cost
- Option C: Simple WhatsApp call link (lowest tech — researcher taps pre-formed WhatsApp link per participant's number)

Recommendation: Start with Option C (WhatsApp call link from admin panel) for the pilot. Upgrade to Option A or B if scaling.

5.15 Support & Emergency Access

UI: Partially Complete | Backend: Not Started | Priority: P1

Components

- Emergency call button — one tap dials the hospital NICU helpline (number per hospital, set in admin panel)
- Contact researcher button — triggers a request to the researcher for a callback
- Community peer posts — read-only view of moderated posts from other mothers (Phase 1: static/curated posts only)

What Needs to Be Built

- Emergency call number configuration in admin panel (per hospital)
- Callback request notification to researcher panel

5.16 Profile & Settings

UI: Complete | Backend: Not Started | Priority: P2

Components

- Mother profile summary — name, phone, hospital, study participant code
- Baby details — name, birth date, gestational age, adjusted age, discharge date
- Activity stats — days active, tasks completed, content viewed, weeks of care
- Badge gallery — earned for consistency milestones (7 days active, first vaccination logged, etc.)
- Settings — language preference, notification preferences, PIN change, logout
- App version and study information

6. Researcher / Admin Panel

The researcher panel is NOT optional for Phase 1. Without it, Purnima Chakrabortty cannot enrol participants, manage the study, access collected data, or conduct follow-ups. It must be built alongside the mother app.

6.1 Access & Authentication

- Separate login from mother app — email + password (researcher only)
- Role-based access: Researcher (full access) | Hospital Nurse (enrolment only)
- Accessible on desktop browser — researcher uses this from a hospital computer

6.2 Participant Management

- Enrol new participant — create account, assign to study or control group, generate participant code
- Participant list — all enrolled mothers with code, site, group, status
- Participant detail view — full profile, all collected data, last activity
- Stratification — filter by birth weight stratum (<1500g / 1500–2500g / >2500g)
- Flag participants who have missed a follow-up or not opened app in 7 days

6.3 Data Collection Access

Researcher can view all submitted data per participant per time point:

- Demographic and infant profile (Tool I)
- Growth measurements and z-scores (Tool II B1)
- TDSC developmental assessment results (Tool II B2)
- Immunization and readmission records (Tool II B3)
- Breastfeeding assessment scores (Tool II B4)
- Knowledge MCQ scores at baseline, 1m, 3m, 6m (Tool III)
- WHO-5 mental well-being scores at all 4 time points (Tool IV)
- PSOC self-efficacy scores at all 4 time points (Tool V)

6.4 Data Export

- Export all participant data as Excel (.xlsx) — one row per participant per time point
- Export format compatible with SPSS and Excel for statistical analysis
- Includes all raw scores, calculated scores, z-scores, and demographic variables
- Blinding-ready: option to export with participant codes only (no names)

6.5 Communication Tools

- Video call initiation — select participant, tap to call (WhatsApp link or embedded video)
- SMS message scheduler — set message content and delivery schedule
- Weekly audio message upload — upload audio file, schedule delivery date
- Bulk notification — send announcement to all active participants

6.6 Hospital Management

- Add hospital — name, code, site supervisor, emergency helpline number
- Assign nurse accounts to hospital
- View per-hospital participant counts and compliance rates

7. Feature Status Summary

Feature	Module	UI Built	Backend	Priority
Authentication (OTP + PIN)	Mother App	Partial	No	P0 — Blocker
Mother & Infant Profile Forms	Mother App	Partial	No	P0 — Blocker
Home Dashboard	Mother App	Complete	No	P0
Daily Care Checklist	Mother App	Complete	No	P0
Knowledge MCQ — Tool III	Research Tool	No	No	P0 — Research
WHO-5 Well-Being — Tool IV	Research Tool	No	No	P0 — Research
PSOC Self-Efficacy — Tool V	Research Tool	No	No	P0 — Research
Growth Tracking + WHO Chart	Mother App	Partial	No	P1
TDSC Developmental Assessment	Research Tool	No	No	P1
Immunization Schedule + Reminders	Mother App	Complete	No	P1
Breastfeeding Log + Assessment	Mother App	No	No	P1
Danger Signs Guide	Mother App	Complete	No	P1
Learning Hub (Content)	Mother App	Partial	No	P1
Daily SMS Messages	Mother App	No	No	P1
Weekly Audio Messages	Mother App	No	No	P1
Telehealth Video Call	Mother App	No	No	P1
Researcher / Admin Panel — All features	Admin Panel	No	No	P1
Bengali Language (full UI)	Mother App	No	No	P1
Offline Mode (Service Worker)	Mother App	No	No	P1
Profile & Settings	Mother App	Complete	No	P2
Badge System	Mother App	Partial	No	P2

8. Key User Flows

8.1 Participant Enrolment Flow (Nurse + Mother at Hospital)

11. Nurse logs into researcher panel on hospital computer or phone
12. Taps 'Enrol New Participant' — enters hospital code
13. Opens SnehoAyu PWA on mother's Android phone
14. Enters mother's phone number — OTP sent
15. Mother enters OTP and creates 4-digit PIN
16. Nurse/mother completes demographic proforma (Tool I) — mother profile + infant profile
17. System assigns participant code and group (study / control) based on randomisation list
18. Baseline data collected — growth measurements, knowledge MCQ, WHO-5, PSOC — all entered by nurse
19. App language confirmed as Bengali
20. 2-hour training session conducted. Nurse demonstrates all main features.
21. Mother goes home. App installed on her phone.
22. Researcher calls mother within 7 days to confirm app use

8.2 Daily Care Flow (Mother at Home)

23. Morning: opens app — Home shows today's progress ring and any new message
24. Reads daily SMS care tip shown as card on Home
25. Taps 'Log Feeding' — enters feed type and count
26. Taps 'Log Temperature' — enters reading; app shows if normal or flagged
27. Throughout day: ticks checklist items as completed
28. Evening: views feeding summary and progress ring update
29. Weekly: opens audio message notification, listens to weekly care talk

8.3 Follow-Up Data Collection Flow (1, 3, 6 Months)

30. Nurse or researcher meets mother at hospital follow-up visit
31. Opens researcher panel and selects participant
32. Takes physical measurements — weight, length, head circumference — enters in app
33. Administers TDSC developmental screening — enters pass/fail per item
34. Reviews immunization record and marks completed vaccines
35. Enters breastfeeding assessment data (Tool II B4)
36. Mother self-completes (or nurse assists) MCQ, WHO-5, and PSOC forms in app
37. All data auto-saved and timestamped
38. Researcher can download updated dataset from admin panel at any time

8.4 Emergency Flow (Mother at Home)

39. Mother notices baby has fast breathing
40. Opens Danger Signs screen (accessible from bottom nav without login)
41. Taps 'Breathing Difficulty' — reads Bengali guidance
42. Taps large red 'Call Hospital' button — phone dialer opens with pre-filled NICU helpline number

9. Technical Requirements

9.1 Frontend

- Framework: React with TypeScript (existing codebase)
- Styling: Tailwind CSS + shadcn/ui component library (existing)
- Routing: React Router with protected routes (existing)
- PWA: manifest.json, service worker (Workbox recommended), installability
- Offline caching: checklist state, danger signs content, last week's messages
- Background sync: data entered offline synced when connection restored
- i18n: react-i18next for Bengali / Hindi / English string management
- Charts: Recharts or Chart.js for WHO growth chart visualisation

9.2 Backend

- REST API — Node.js (Express) or Python (FastAPI) recommended
- Database: PostgreSQL — structured relational data required for research dataset integrity
- Authentication: JWT with refresh tokens; OTP via SMS gateway
- File storage: AWS S3 or equivalent for audio message files
- SMS gateway: MSG91 (India) for OTP and daily messages
- Video calls: WhatsApp deep link (Phase 1) or Daily.co SDK
- Multi-tenancy: all data scoped per hospital and per participant

9.3 Hosting

- Cloud hosting: AWS, GCP, or Azure — India region preferred for data residency
- Separate environments: Development | Staging | Production
- HTTPS enforced — required for PWA service worker
- Automated backups: daily database backup retained for 30 days

9.4 Performance

- First Contentful Paint: under 3 seconds on 4G
- Lighthouse PWA score: > 90
- Total initial bundle: < 500KB gzipped
- All images served as WebP with responsive sizing

9.5 Browser Support

- Android Chrome 90+ — primary target
- Samsung Internet 14+ — secondary
- Safari iOS 15+ — tertiary
- No Internet Explorer support required

10. Non-Functional Requirements

10.1 Security

- All API endpoints require authentication — no unauthenticated data access
- Research data encrypted at rest (AES-256) and in transit (TLS 1.3)
- Participant codes used in data export — no names in exported research dataset
- OWASP Top 10 review required before production launch
- No third-party analytics that collect health data (no Google Analytics on research screens)

10.2 Data Privacy — DPDPA 2023

- Explicit informed consent collected digitally at account creation
- Participants can request deletion of personal data at any time
- Research data retained for minimum 5 years post-study (institutional requirement)
- Data shared with hospital partners is anonymised (participant codes only)

10.3 Availability

- Target uptime: 99.5% for all mother-facing API endpoints
- Maintenance windows: 2 AM – 4 AM IST only (minimum disruption)
- Graceful degradation: if API is down, app must still work offline

10.4 Accessibility

- WCAG 2.1 AA compliance for all core screens
- Minimum touch target: 48×48px (critical for anxious parents with shaking hands)
- Font size: minimum 16px for body text
- Colour contrast: minimum 4.5:1 for all text
- Audio messages must have a transcript available (accessibility + literacy)

11. Build Phases & Delivery Plan

Total estimated build: 18 weeks from project kickoff to pilot launch. The pilot at Deban Mahato MCH, Purulia must be completed before main study enrolment begins at the three primary sites.

Phase	Duration	Deliverables
Phase 1A Foundation	Weeks 1–4	Real OTP authentication + PIN login. Mother and infant profile forms (Tool I). Backend API scaffold with all data models. Researcher panel: login, enrolment, participant list.
Phase 1B Core Care + Research Tools	Weeks 5–9	Home dashboard wired to live data. Daily checklist with persistence. Knowledge MCQ (Tool III) in-app. WHO-5 Well-Being (Tool IV) in-app. PSOC Self-Efficacy (Tool V) in-app. All 3 research forms with scoring, history, and export.
Phase 1C Health Tracking	Weeks 10–13	Growth tracking with WHO chart and z-scores. TDSC developmental assessment. Immunization schedule + mark-as-done. Breastfeeding log + assessment form. Danger signs with live emergency call.
Phase 1D Communication + Content	Weeks 14–16	Daily SMS integration (MSG91). Weekly audio message delivery. Telehealth video call (WhatsApp link). Learning hub with Bengali content. Support screen with emergency call and callback request.
Phase 1E PWA + Localisation + Polish	Weeks 17–18	Full Bengali UI (all strings). Service worker + offline mode. PWA manifest + installability. Performance audit. Security review. Researcher panel: data export to Excel/SPSS. Pilot launch at Purulia.

11.1 Post-Pilot

- Pilot runs for 4–6 weeks at Deban Mahato MCH with ~20 participants
- Researcher and developer review: bugs, UX issues, missing data fields
- Bengali content review and corrections
- Main study enrolment begins at Bankura, Burdwan, Purba Bardhaman

12. Open Questions for Client Decision

#	Question	Why It Matters	Owner	Status
1	Should the PSOC and WHO-5 forms be self-completed by the mother or entered by the nurse at the follow-up visit?	Affects UI design — self-complete needs simpler Bengali UX; nurse-entry can be more clinical.	Purnima Chakraborty	Open
2	Is randomisation to study/control group done manually by the researcher before enrolment, or should the app handle it automatically?	Determines complexity of enrolment flow in admin panel.	Purnima Chakraborty	Open
3	What is the approved Bengali translation for all 15 MCQ questions and the PSOC items? Who validates it?	Cannot build Bengali forms without validated translations.	Purnima Chakraborty + Language Expert	Open
4	Who will create the weekly audio message content and record the audio files?	Must be scheduled into study preparation timeline.	Purnima Chakraborty	Open
5	Who manages the daily SMS content library (180+ messages for 6 months)?	Large content task — needs assignment before dev starts.	Purnima Chakraborty	Open
6	Should the telehealth be WhatsApp-based (simple, no dev) or an in-app video call?	Affects development time and cost significantly.	Purnima Chakraborty	Open
7	Is DPDPA 2023 (India data privacy law) compliance formally required by the hospital IRBs?	Legal requirement may affect data storage architecture.	Institutional IRB	Open
8	What statistical software will be used for analysis — SPSS or R? What exact column format is needed for export?	Data export format must match analysis software expectations.	Statistician	Open
9	Should the control group mothers have any access to the app? (The proposal says standard care only — confirming no app access.)	Defines whether one codebase or two apps are needed.	Purnima Chakraborty	Open

13. Existing Codebase Reference

The following components are already built in the current frontend. Developers can treat these as the base to extend.

Area	Files / Components	Notes
Routing & Auth	src/App.tsx, src/lib/auth.tsx, src/lib/use-auth.ts	Auth is localStorage stub — replace with real OTP + JWT backend
Pages	Login, Index, Checklist, Growth, Learn, Support, DangerSigns, Immunization, Profile	All pages exist; all data is mocked
Dashboard Components	src/components/dashboard/*	UI complete; wire to backend API
Layout	AppLayout, DesktopSidebar, MobileHeader, MobileSidebar, BottomNavigation	Complete — keep as is
UI Library	src/components/ui/* (shadcn/ui primitives)	Complete — reuse across all new screens
Toasts & Hooks	src/hooks/use-toast.ts, use-mobile.ts	Complete
Not yet built	TDSC screen, Breastfeeding screen, MCQ form, WHO-5 form, PSOC form, Researcher panel, Audio player, SMS history	Build from scratch using existing UI library

Note for developers: The UI shell (navigation, layout, component library, routing) is production-ready. The primary work remaining is: (1) building the missing feature screens, (2) connecting all screens to a real backend API, and (3) implementing Bengali internationalisation.

14. Document Sign-Off

Role	Name
Research	Purnima Chakrabortty
Lead Developer	Antik Mondal
Project Manager	Antik Mondal